

A Generative Reconstruction of Low- ℓ CMB B-Mode Signal using Reverse Diffusion in Deep Learning arXiv:2512.22683

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Abstract

Detecting primordial B-mode polarization of the Cosmic Microwave Background (CMB) provides a direct probe of inflationary gravitational waves. However, the signal is extremely faint and contaminated by gravitational lensing, instrumental noise, and astrophysical foregrounds. Here we present a score-based diffusion approach, formulated using variance-exploding stochastic differential equations (VE-SDEs), to reconstruct the primordial B-mode angular power spectrum from contaminated observations. The method employs a reverse SDE guided by a score model trained exclusively on random realizations of the primordial low ℓ B-mode angular power spectrum corresponding to a fixed tensor-to-scalar ratio $r = 0.001$.

During inference, the reverse SDE iteratively drives the observed angular power spectrum toward the learned primordial manifold, effectively denoising and delensing the input. The model is tested on simulated observational spectra that include gravitational lensing, complex polarized foreground combinations, and instrumental noise characteristics representative of the proposed ECHO mission. The trained score model learns the underlying statistical distribution of the primordial B-mode field for the given r , which acts as a physics-guided prior that can generate new, consistent realizations of the signal. This approach provides a robust framework for primordial signal recovery in future CMB polarization missions.

Formalism

Forward and reverse SDEs

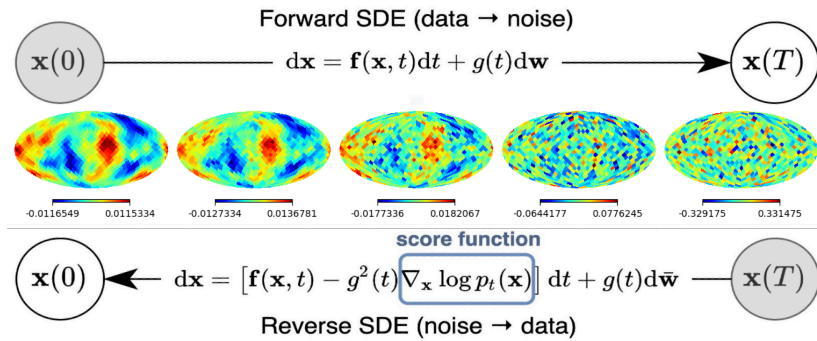


Figure: Top panel shows transformation of the data to a noise dominated distribution by slowly adding noise to it using a forward SDE. The bottom panel shows that samples of noise minimized data can be generated by using the reverse SDE method. In both cases the differential equations used are mentioned. The reverse SDE computation requires knowledge of score function.

Forward SDE

$$d\mathbf{x} = f(\mathbf{x}, t)dt + g(\mathbf{x}, t)d\mathbf{w}. \quad (1)$$

Reverse SDE

$$d\mathbf{x} = [f(\mathbf{x}, t) - g(\mathbf{x}, t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] dt + g(\mathbf{x}, t)d\bar{\mathbf{w}}. \quad (2)$$

The variance–exploding SDE (VE-SDE)

$$f(\mathbf{x}, t) = 0, \quad g(t) = \sigma(t) \sqrt{2 \ln \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)}. \quad (3)$$

with an exponential noise schedule

$$\sigma(t) = \sigma_{\min} \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^t, \quad t \in [0, 1]. \quad (4)$$

Formalism

Score network and denoising score matching

We approximate $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ using deep learning model $s_\theta(\mathbf{x}_t, t)$, referred to as **ScoreNet1D**.

We employ a variance-exploding SDE, which induces a Gaussian perturbation of the form

$$\mathbf{x}_t = \mathbf{x}_0 + \sigma_{\text{marg}}(t)\mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (5)$$

The corresponding conditional score takes the form

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{x}_0) = -\frac{\mathbf{x}_t - \mathbf{x}_0}{\sigma_{\text{marg}}(t)^2} = -\frac{\mathbf{z}}{\sigma_{\text{marg}}(t)}. \quad (6)$$

$$\mathcal{L}_{\text{WDSM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{z}} \times \left[\lambda(t) \left\| s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | \mathbf{x}_0) \right\|^2 \right]. \quad (7)$$

$$\mathcal{L}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{z}} \left[\left\| \sigma_{\text{marg}}(t) s_\theta(\mathbf{x}_t, t) + \mathbf{z} \right\|^2 \right]. \quad (8)$$

Formalism

Euler–Maruyama discretization

$$\Delta \mathbf{w}_k = \mathbf{w}_{t_{n+1}} - \mathbf{w}_{t_n} \sim \sqrt{\Delta t} \xi_k, \quad \xi_k \sim \mathcal{N}(0, 1). \quad (9)$$

Forward SDE

$$\mathbf{x}_{k+1} = \mathbf{x}_k + g(t_k) \Delta \mathbf{w}_k = \mathbf{x}_k + \sigma(t_k) \sqrt{2 \ln \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)} \sqrt{\Delta t} \boldsymbol{\xi}_k, \quad (10)$$

Reverse SDE

$$\begin{aligned} \mathbf{x}_{k+1} = \mathbf{x}_k &- 2\sigma(t_k)^2 \ln \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right) \nabla_{\mathbf{x}} \log p_{t_k}(\mathbf{x}_k) \Delta t \\ &+ \sigma(t_k) \sqrt{2 \ln \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)} \sqrt{\Delta t} \boldsymbol{\xi}_k. \end{aligned} \quad (11)$$

Simulations

Training Data: Primordial CMB B -mode Realizations

We generate the theoretical primordial B -mode angular power spectrum $C_\ell^{BB(\text{tensor})}$ using the CAMB Boltzmann code. HEALPix (Hierarchical Equal Area iso-Latitude Pixelization) is then used to simulate full-sky Stokes- Q/U maps from these spectra for a large number of independent random seeds. The transfer function

$$b_\ell = \exp \left[-\frac{\ell(\ell+1)\sigma_b^2}{2} \right]. \quad (12)$$

The Gaussian width σ_b

$$\sigma_b = \frac{\theta_{\text{FWHM}}}{\sqrt{8 \ln 2}}, \quad (13)$$

where we adopt the 28 GHz channel specification of the proposed ECHO mission, corresponding to $\theta_{\text{FWHM}} = 24.8'$.

Simulations

Testing Data: Realistic Observational Spectra

$$\hat{C}_\ell^{BB(\text{obs})} = \left(\hat{C}_\ell^{BB(\text{tensor})} + C_\ell^{BB(\text{lensing})} + F_\ell^{BB} + \hat{N}_\ell^{BB} \right) B_\ell^2. \quad (14)$$

1. Polarized foreground contamination

We model polarized Galactic foreground emission at 28 GHz using the PySM3 sky simulation framework

$$\begin{aligned} \text{F1} : (a2, s1, d1), & \quad \text{F2} : (a2, s1, d4), \\ \text{F3} : (a2, s3, d1), & \quad \text{F4} : (a2, s3, d4). \end{aligned}$$

2. Instrumental noise

$$\sigma_v^2 = \left(\frac{\sigma_{\text{rms}} \pi}{60 \times 180} \right)^2 \left(\frac{N_{\text{pix}}}{4\pi} \right), \quad (15)$$

and Gaussian-distributed Q/U noise maps are generated as

$$Q_{\text{noise}}(\hat{n}), U_{\text{noise}}(\hat{n}) \sim \mathcal{N}(0, \sigma_v^2).$$

Model Training

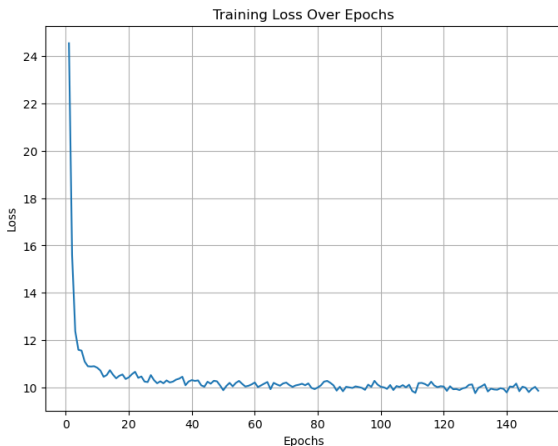


Figure: Training loss evolution for **ScoreNet1D** under the weighted denoising score-matching objective over 150 epochs. The loss decreases rapidly during the initial phase and saturates after ~ 100 epochs, indicating stable convergence of the learned score estimator.

Reconstruction Results and Statistical Validation

Reconstruction of Individual Realizations

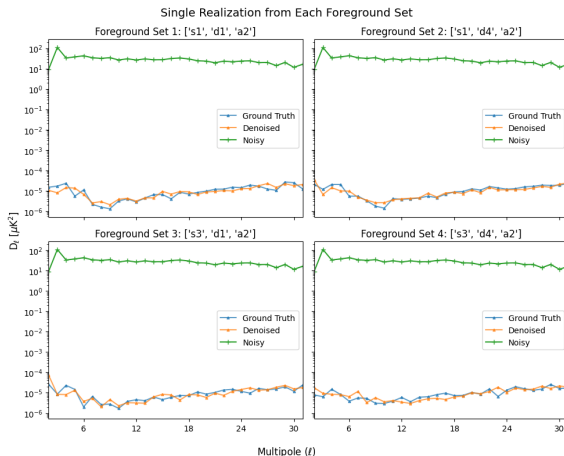


Figure: Reconstruction of a single low- ℓ CMB B -mode realization for four distinct foreground configurations (F1–F4).

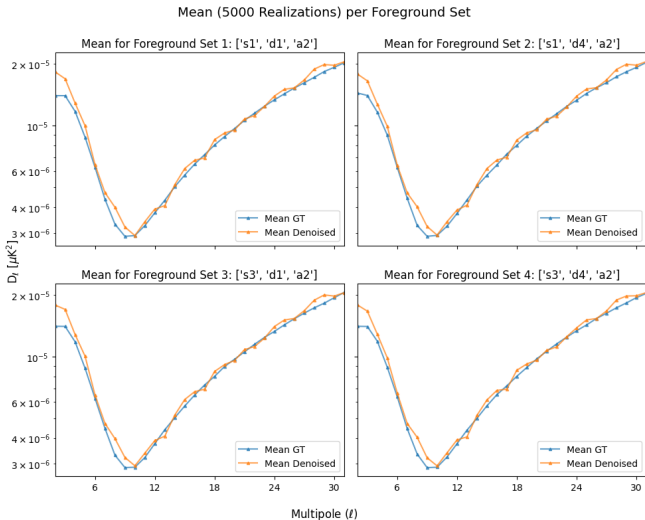


Figure: Ensemble-averaged primordial B -mode power spectrum obtained from 5,000 independent realizations for each foreground configuration.

Reconstruction Results and Statistical Validation

Reconstruction Uncertainty and Cosmic Variance Consistency

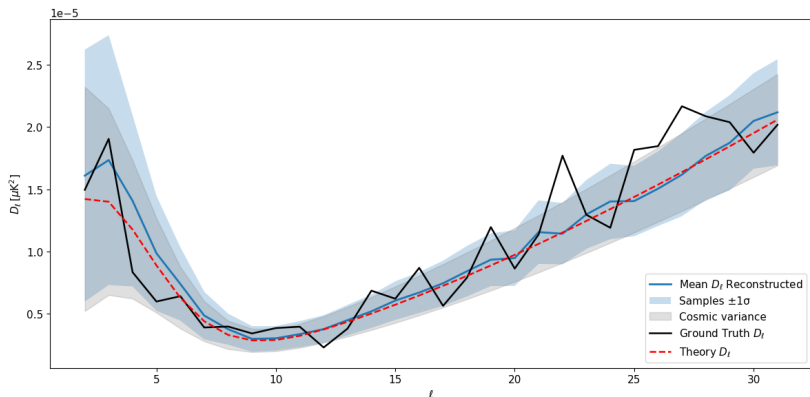


Figure: Ensemble-averaged reconstruction of the primordial B -mode D_ℓ spectrum from 10,000 independent reverse VE-SDE samples starting from the same observed realization.

Reconstruction Results and Statistical Validation

Score field geometry validation

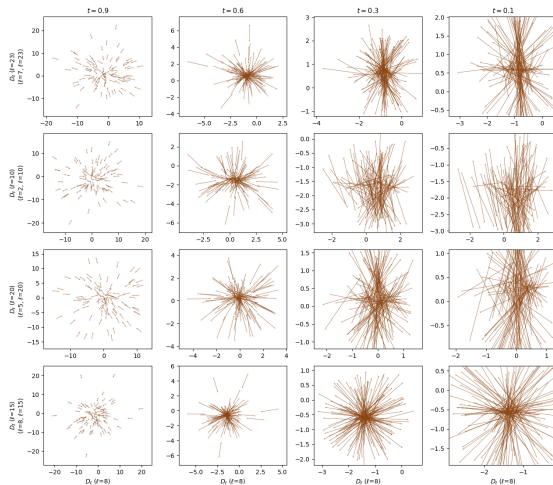


Figure: Visualization of the learned score field in D_ℓ space at multiple diffusion times for selected multipole pairs.

Reconstruction Results and Statistical Validation

Likelihood and Distribution Agreement

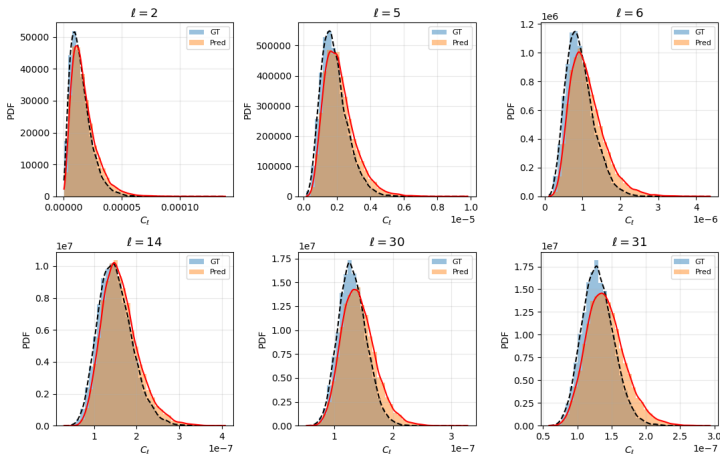


Figure: These tail differences dominate the Kullback–Leibler (KL) contribution and do not indicate a systematic bias in the bulk of the distribution.

Summary and Conclusion

In this work, we developed a score-based stochastic differential equation (SDE) framework that performs primordial B -mode reconstruction through a learned reverse diffusion process in the low-multipole regime. By training a score model on simulated primordial spectra at $r = 0.001$, the method learns the underlying data distribution and uses this learned score to iteratively remove noise, lensing contamination, and polarized foregrounds through the reverse VE-SDE. Because the model is trained only on primordial spectra and not on any specific foreground templates, the reconstruction is driven by the learned structure of the primordial signal rather than by assumptions about the nature of contaminants.

Looking ahead, this work provides a foundation for developing more extensive models that learn the continuous family of primordial spectra over a range of r values rather than a fixed amplitude. Such models could offer a robust, data-driven mechanism for probing the primordial gravitational-wave background and may eventually serve as state-of-the-art simulators for next-generation missions. With further refinement and broader training datasets, the score-based SDE framework has the potential to become a widely applicable tool for precision reconstruction of the primordial CMB B -mode signal.